

Data Communication & Computer Networks

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Road Map

1 Network Classification

Based on area

Based on component role

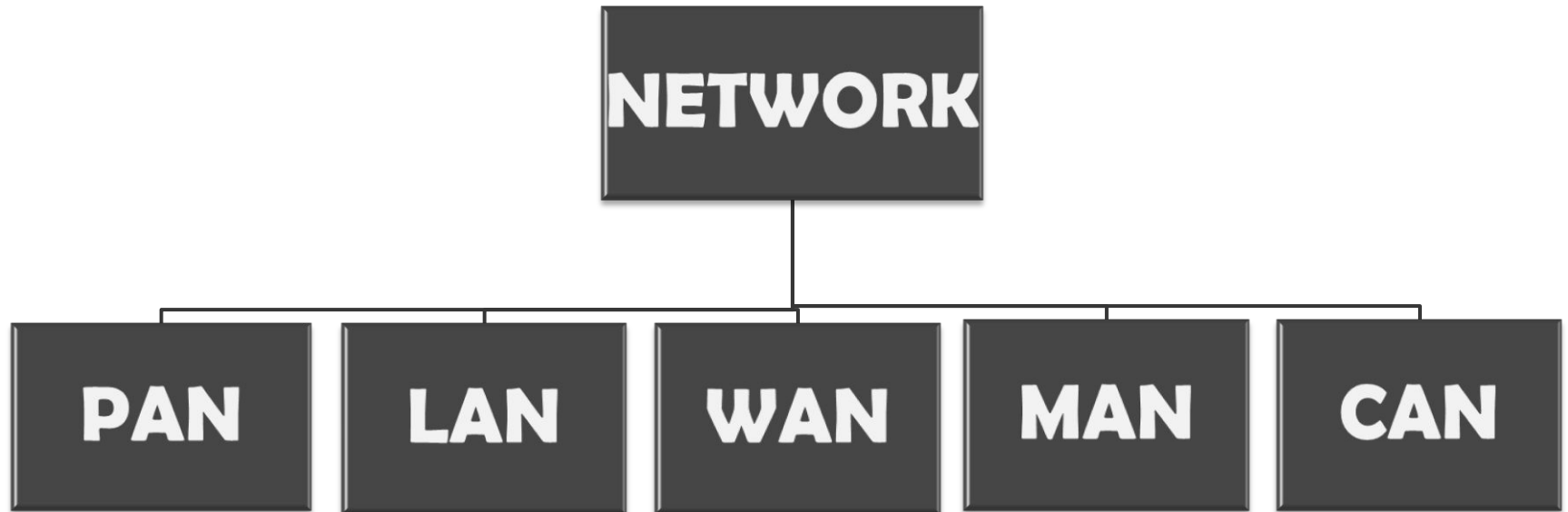
Based on connectivity

Based on transmission medium

2 Internet of Things

LPWAN

NETWORK CLASSIFICATION OF AREA BY THEIR GEOGRAPHY



LOCAL AREA NETWORK(LAN)

- LAN is a network which is designed to operate over a small physical area such as an office, factory or a group of buildings.
- LAN's are easy to design and troubleshoot
- Exchange of information and sharing of resources becomes easy because of LAN.
- In LAN all machines are connected to a single cable.
- Different types of topologies such as star, tree, bus, ring, etc Can be used
- It is usually a privately owned network.

WIDE AREA NETWORK(WAN)

- When network spans over a large distance or when the computers to be connected to each other are at widely separated locations a local area network cannot be used. A wide area network(WAN) is installed.
- The communication between different users of WAN is established using leased telephone lines, satellite links and similar channels.
- It is cheaper and more efficient to use the phone network for the link.
- Most WAN networks are used to transfer large blocks of data between its users.

PERSONAL AREA NETWORK(PAN)

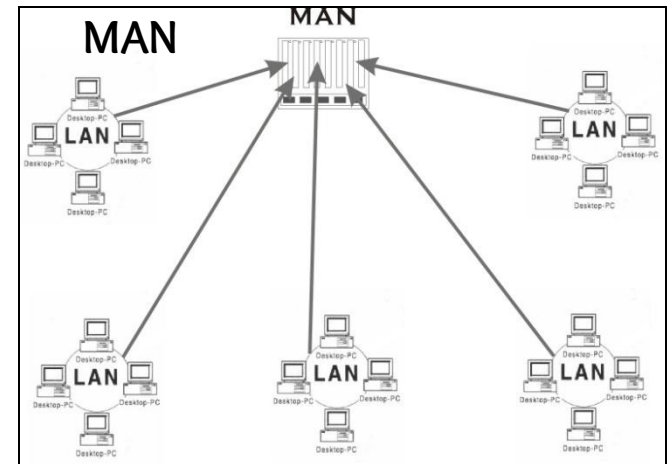
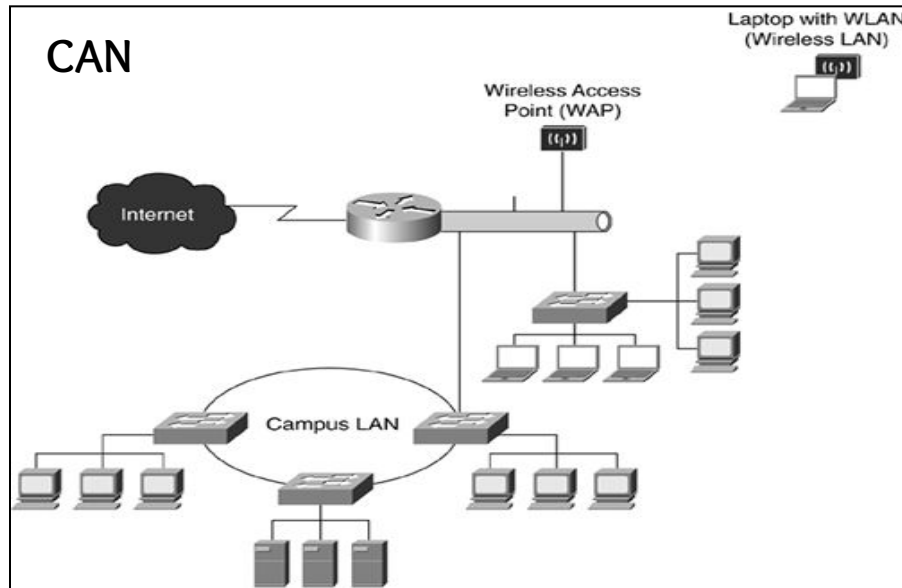
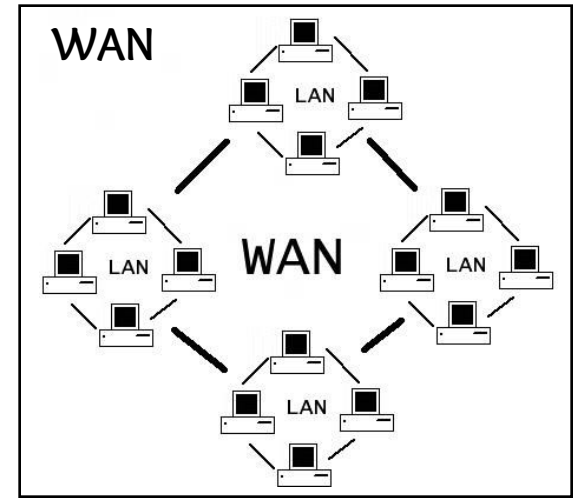
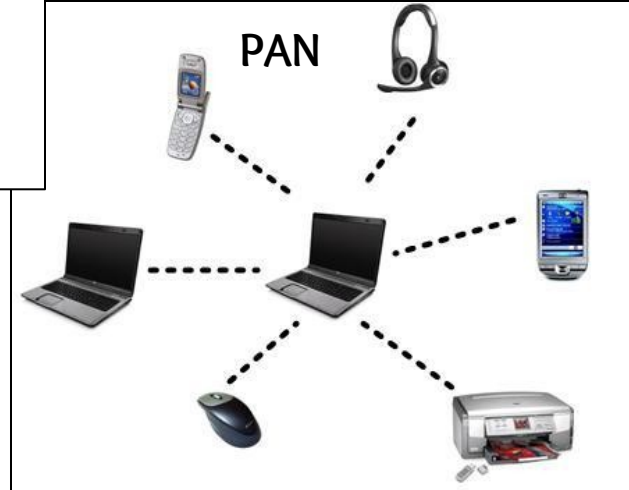
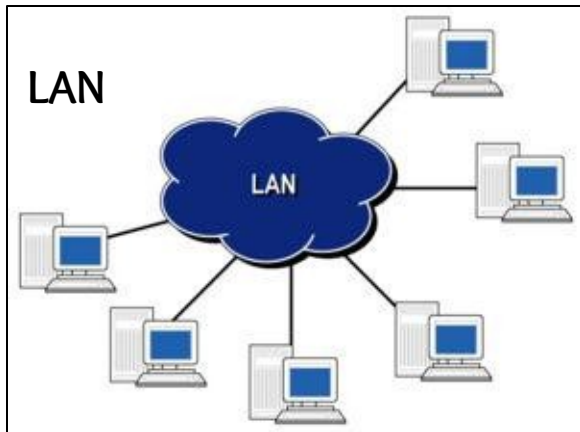
- ❑ A personal area network is a computer network organized around an individual person.
- ❑ It generally consists of a mobile computer, a cell phone or personal digital assistant. PAN enables the communication among these devices.
- ❑ The PANs can be constructed using wireless or cables.

CAMPUS AREA NETWORK(CAN)

- The campus area network is made up of an interconnection of LAN with limited geographical area.
- Network equipment's such as switches, routers and the transmission media i.e. optical fibre etc are almost entirely owned by the campus owner.

METROPOLITAN AREA NETWORK(MAN)

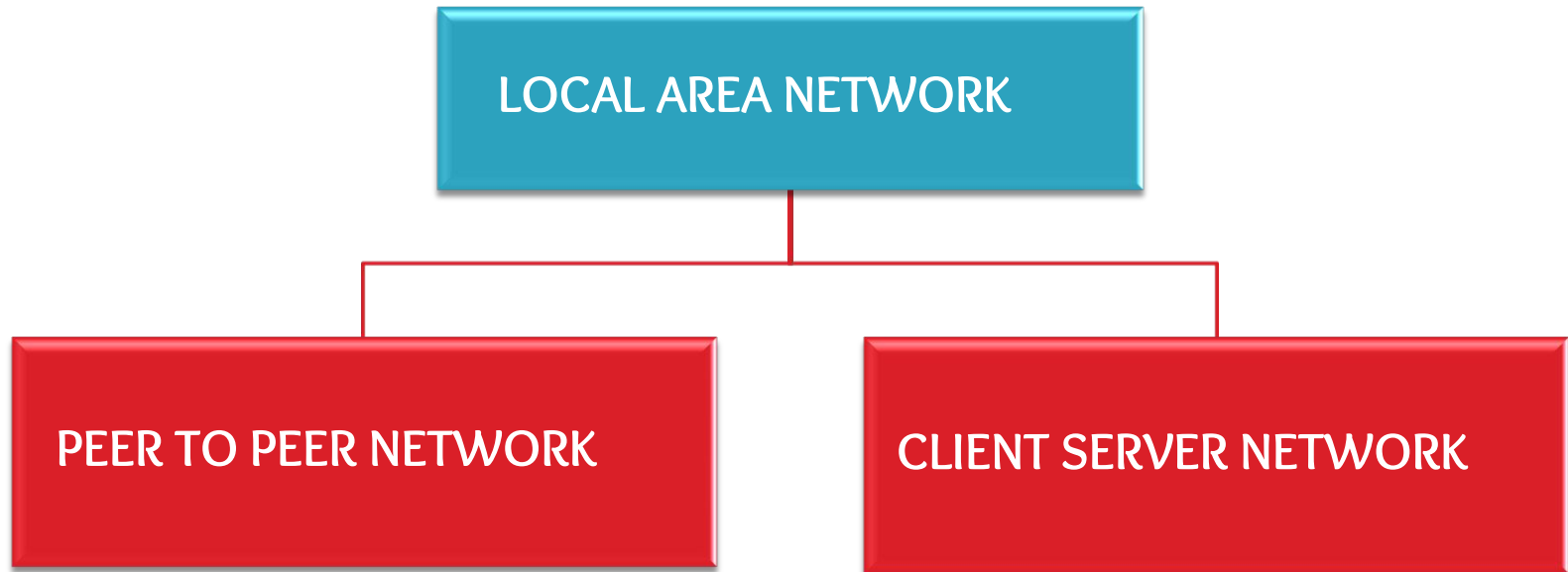
- It is in between LAN & WAN technology that covers the entire city.
- It uses similar technology as LAN.
- It can be a single network such as cable TV network



DISTINGUISH BETWEEN LAN,WAN,MAN

PARAMETERS	LAN	WAN	MAN
Ownership of network	Private	Private or public	Private or public
Geographical area covered	Small	Very large	Moderate
Design and maintenance	Easy	Not easy	Not easy
Communication medium	Coaxial cable	PSTN or satellite links	Coaxial cables, PSTN, optical fibre, cables, wireless
Bandwidth	Low	High	moderate
Data rates(speed)	High	Low	moderate

NETWORK CLASSIFICATION BY THEIR COMPONENT ROLE



PEER TO PEER NETWORK

- In peer to peer network each computer is responsible for making its own resources available to other computers on the network.
- Each computer is responsible for setting up and maintaining its own security for these resources.
- Also each computer is responsible for accessing the required network resources from peer to peer relationships.
- Peer to peer network is useful for a small network containing less than 10 computers on a single LAN .
- Peer to peer networks do not have a central control system.
There are no servers in peer networks.
- Peer networks are amplified into home group.

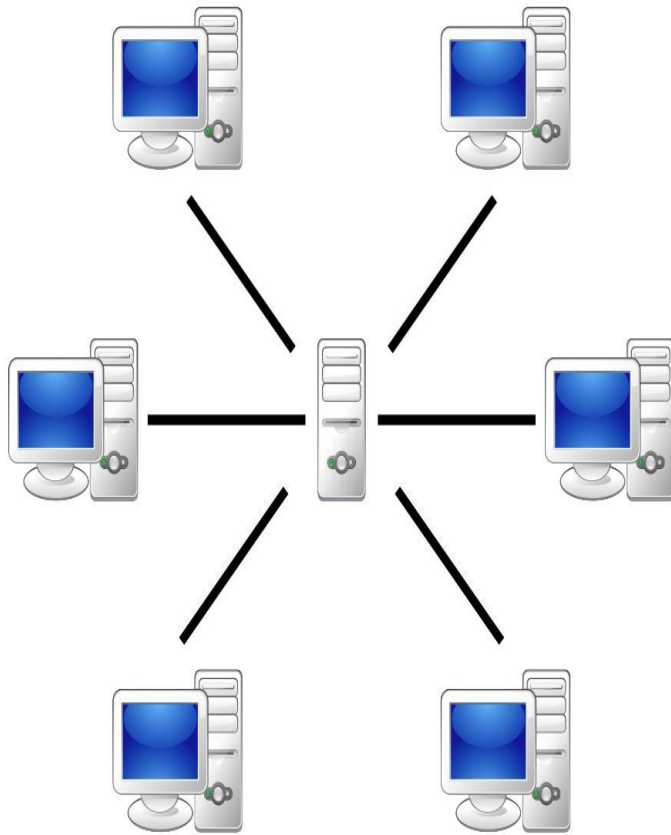
ADVANTAGES & DISADVANTAGES OF PEER TO PEER NETWORK

Advantages:

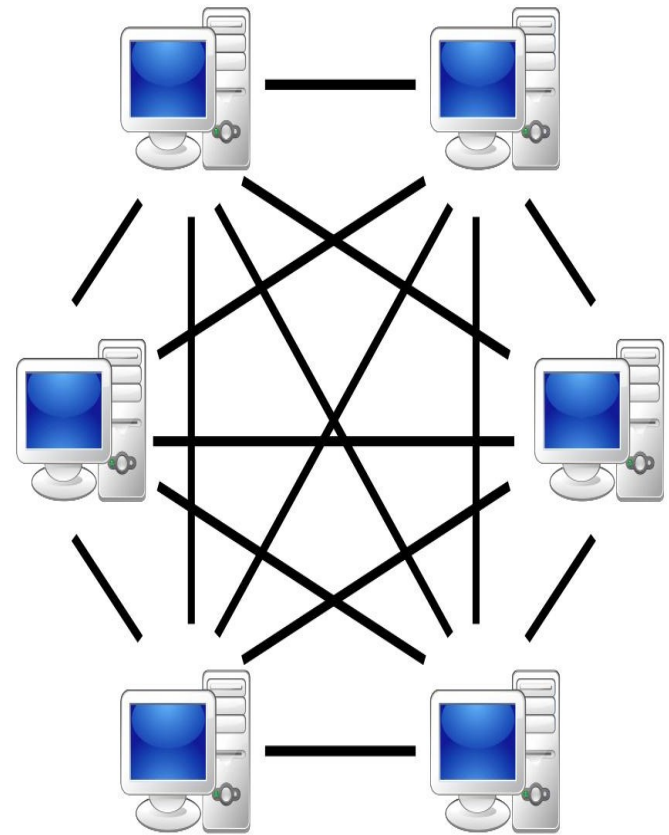
- Use less expensive computer hardware
- Easy to administer
- Easy setup & low cost

Disadvantages:

- Not very secure
- No central point of storage or file archiving
- Additional load on computer because of resource sharing



Server-based



P2P-network

CLIENT/SERVER NETWORK

- In client-server network relationships, **certain computers act as server and other act as clients. A server is simply a computer, that available the network resources and provides service to other computers when they request it. A client is the computer running a program that requests the service from a server.**
- **Local area network(LAN) is based on client server network relationship.**
- **A client-server network is one in which all available network resources such as **files, directories, applications and shared devices**, are centrally managed and hosted and then are accessed by client.**
- **Client server network are defined by the presence of servers on a network that provide security and administration of the network.**

ADVANTAGES AND DISADVANTAGES OF CLIENT-SERVER NETWORK

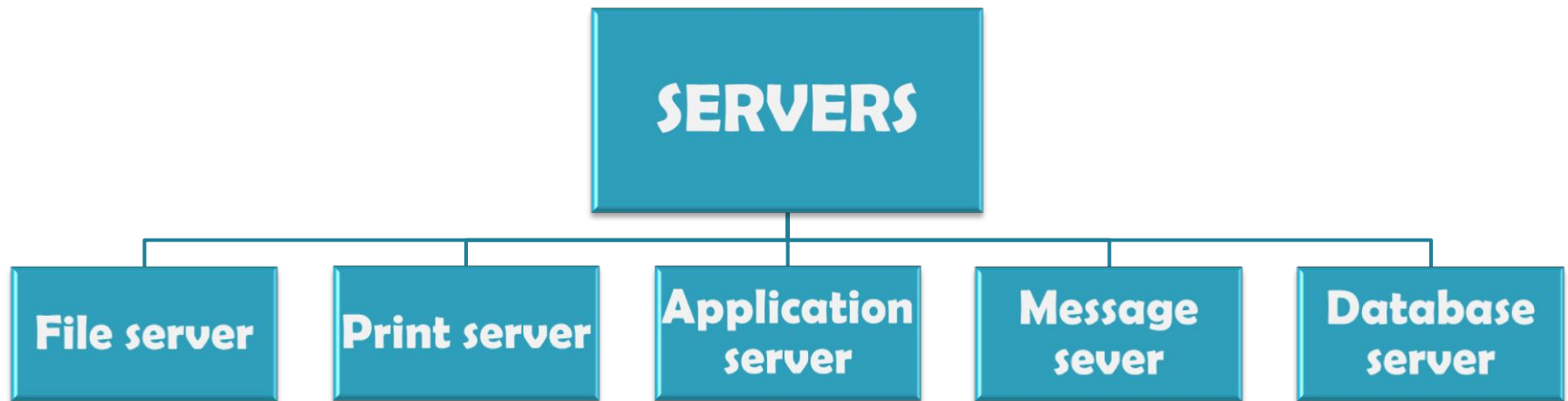
Advantages:

- Very secure
- Better performance
- Centralized backup
- very reliable

❖ Disadvantages:

- requires professional administration
- More hardware-intensive
- More software intensive
- Expensive dedicated software

TYPES OF SERVERS



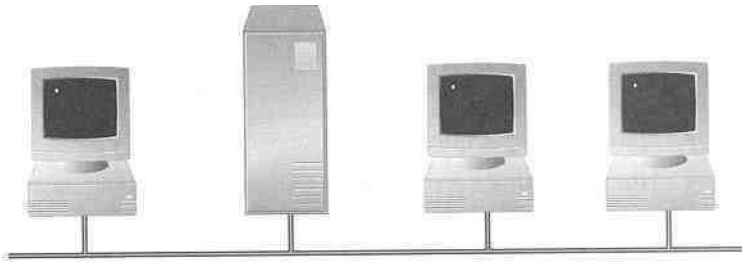
TYPES OF SERVERS

- **File server:** These servers provide the services for storing, retrieving and moving the data. A user can read, write, exchange and manage the files with the help of file servers.
- **Printer server:** The printer server is used for controlling and managing printing on the network. It also offers the fax service to the network users.
- **Application server:** The expensive software and additional computing power can be shared by the computers in a network with the help of application servers.
- **Message server:** It is used to co-ordinate the interaction between users, documents and applications. The data can be used in the form of audio, video, binary, text or graphics.
- **Database server:** It is a type of application server. It allows the users to access the centralised strong database.

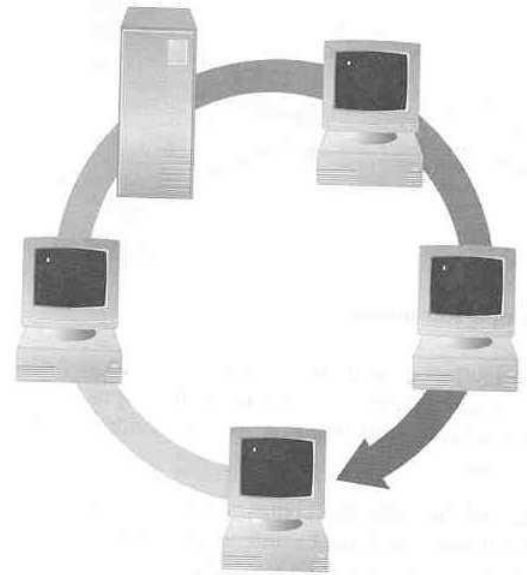
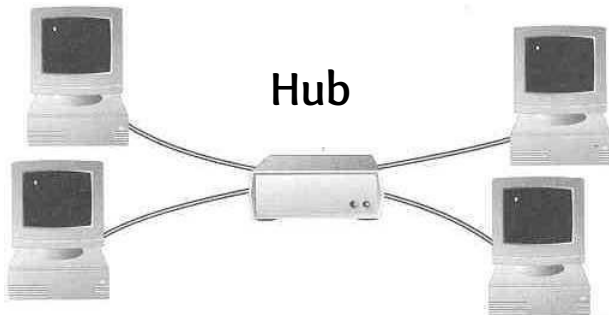
Network Classification based on TOPOLOGY(Connectivity)

- How so many computers are connected together?

Bus Topology Ring Topology



Star Topology



TOPOLOGY

- Bus Topology

- Simple and low-cost
- A single cable called a **trunk** (**backbone**, **segment**)
- Only one computer can send messages at a time
- Computer only listen for, not regenerate data

- Star Topology

- Each computer has a cable connected to a single point
- More cabling, hence **higher cost**
- All signals transmission through the hub; **if down, entire network down**
- Depending on the intelligence of hub, two or more computers may send message at the same time

TOPOLOGY

- **Ring Topology**

- Every computer serves as a repeater to boost signals
- Typical way to send data:
 - Token passing
 - only the computer who gets the token can send data
- Disadvantages
 - Difficult to add computers
 - More expensive
 - If one computer fails, whole network fails

Network Classification based on TRANSMISSION MEDIA

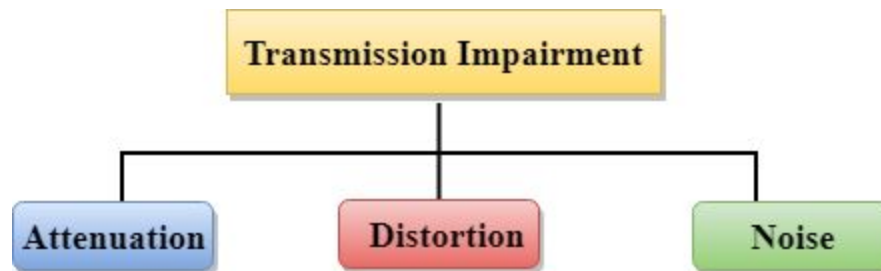
- ❑ **Two main categories:**
 - ❑ **Guided** — wires, cables
 - ❑ **Unguided** — wireless transmission, e.g. radio, microwave, infrared, sound, sonar
- ❑ **We will concentrate on guided media here:**
 - ❑ **Twisted-Pair cables:**
 - ❑ **Unshielded Twisted-Pair (UTP) cables**
 - ❑ **Shielded Twisted-Pair (STP) cables**
 - ❑ **Coaxial cables**
 - ❑ **Fiber-optic cables**

Some factors need to be considered for designing the transmission media:

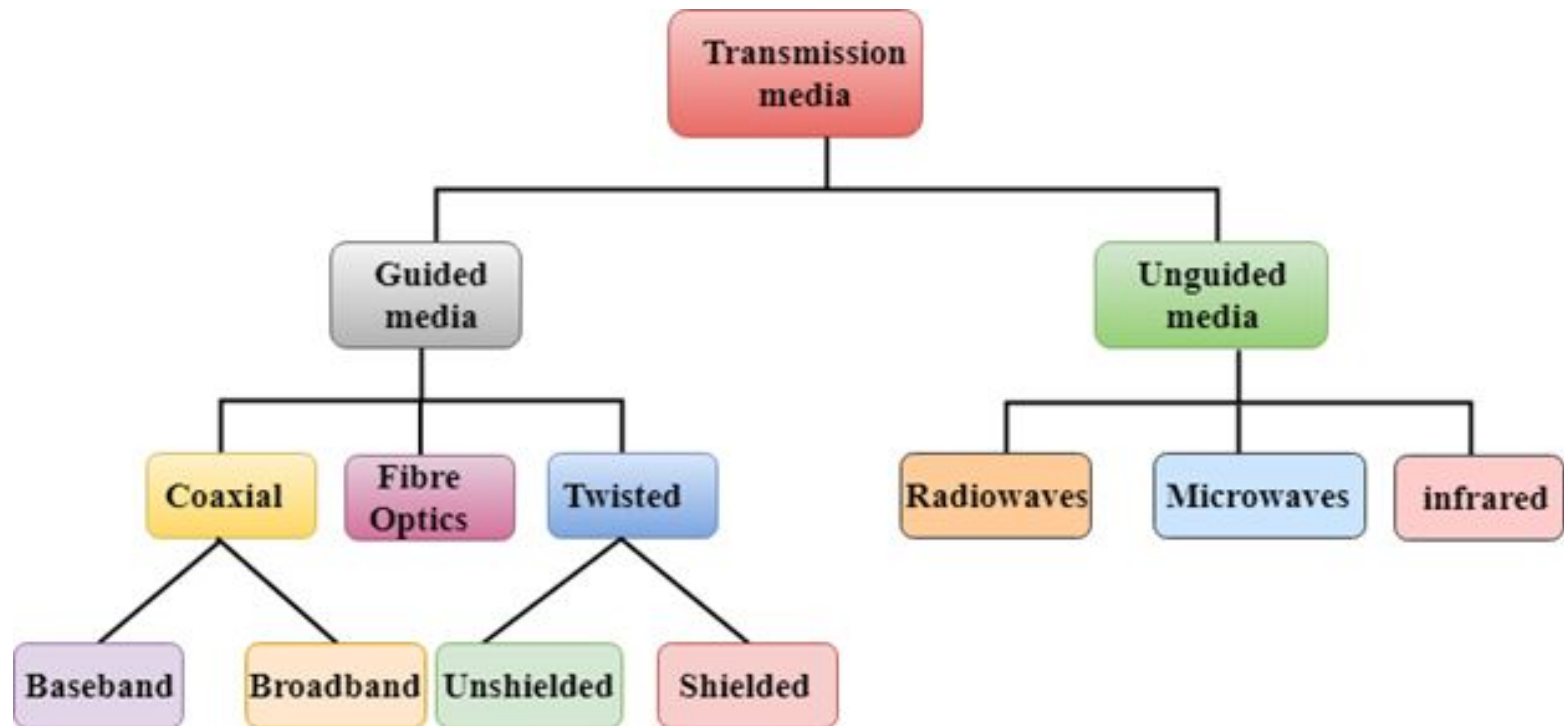
- ❖ **Bandwidth:** All the factors are remaining constant, the greater the bandwidth of a medium, the higher the data transmission rate of a signal.
- ❖ **Transmission impairment:** When the received signal is not identical to the transmitted one due to the transmission impairment. The quality of the signals will get destroyed due to transmission impairment.
- ❖ **Interference:** An interference is defined as the process of disrupting a signal when it travels over a communication medium on the addition of some unwanted signal.

Causes Of Transmission Impairment:

- ❖ **Attenuation:** Attenuation means the loss of energy, i.e., the strength of the signal decreases with increasing the distance which causes the loss of energy.
- ❖ **Distortion:** Distortion occurs when there is a change in the shape of the signal. This type of distortion is examined from different signals having different frequencies. Each frequency component has its own propagation speed, so they reach at a different time which leads to the delay distortion.
- ❖ **Noise:** When data is travelled over a transmission medium, some unwanted signal is added to it which creates the noise.



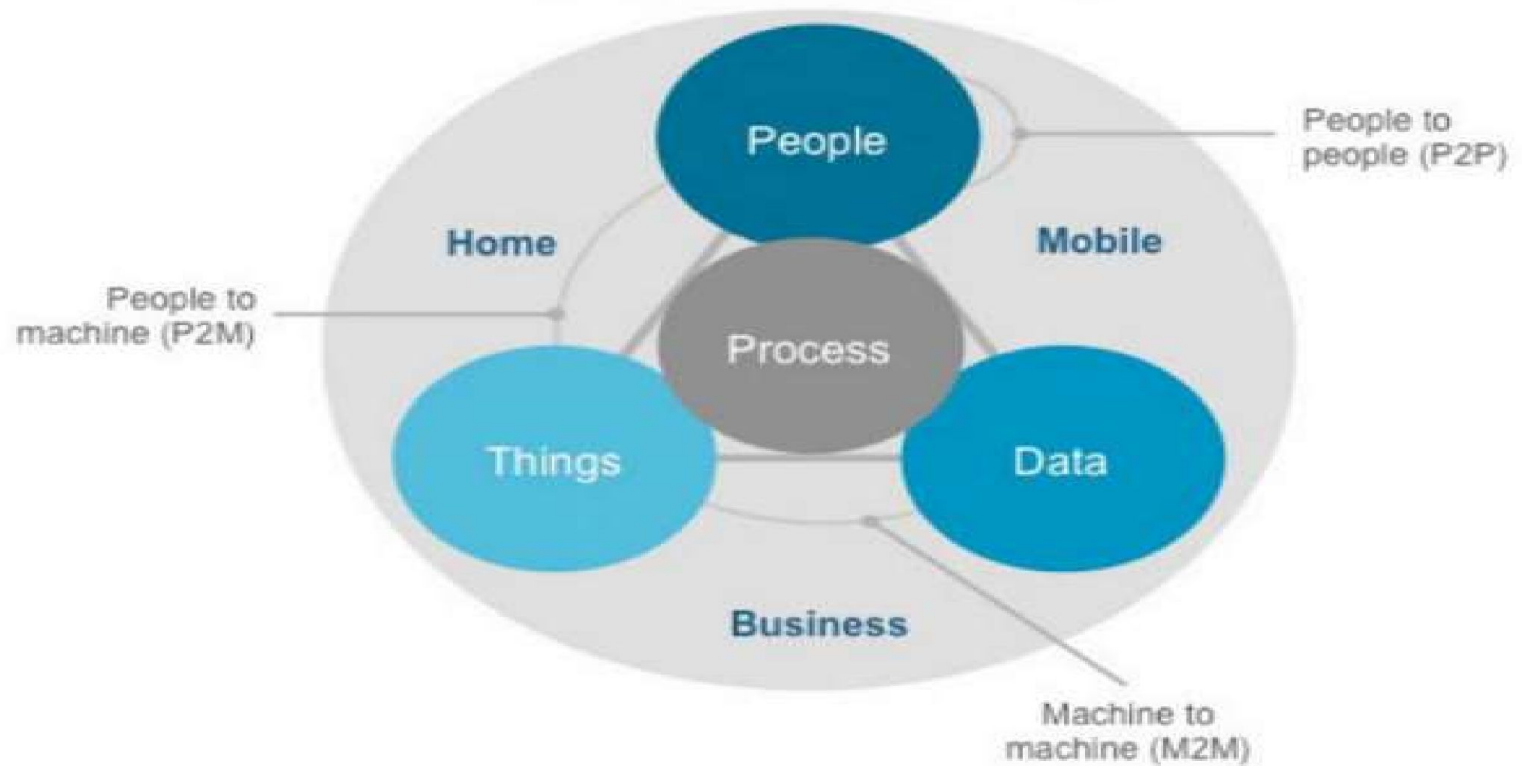
Classification Of Transmission Media:



Internet of Things (IoT)

- The **Internet of Things (IoT)** is the network of physical objects—devices, vehicles, buildings and other items embedded with electronics, software, sensors and connectivity that enables ~~network~~ objects to collect and exchange data.
- Allows objects to be sensed and remotely across existing controlled infrastructure network

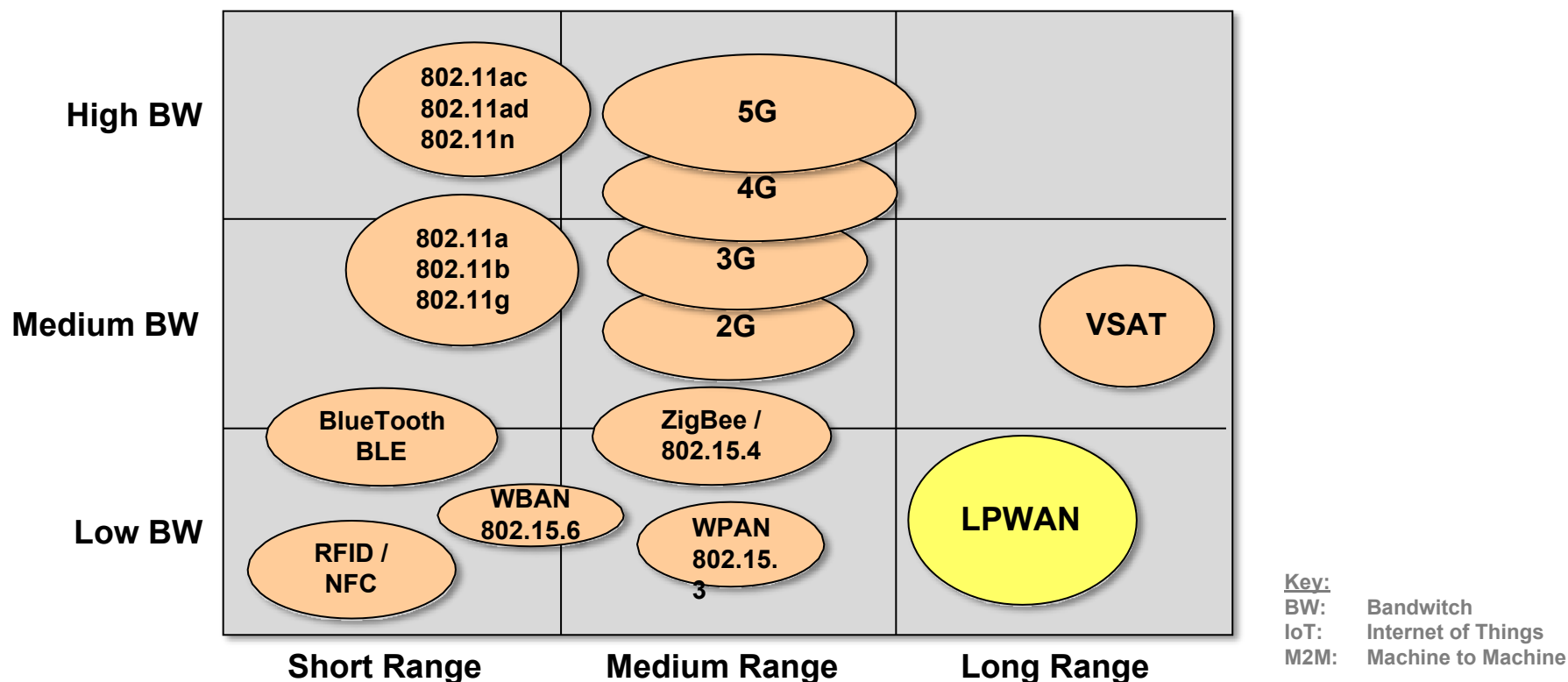
Internet of Everything



LPWAN – Low Power Wide Area Network

Different wireless technologies cover different applications with regard to range and bandwidth. Long-range applications with low bandwidth requirements that are typical for IoT and M2M scenarios are not well supported by these existing technologies.

LPWAN technologies are targeted at these emerging applications and markets.



Q/A